Exercise J-M

1. The equation $x^2 - 4x + [x] + 3 = x[x]$, where [x] denotes the greatest integer function, has : **[JEE(Main)Jan.2023]**

(A) no solution (B) a unique solution in $(-\infty, \infty)$

(C) a unique solution in $(-\infty, 1)$ (D) exactly two solution in $(-\infty, \infty)$

समीकरण $x^2 - 4x + [x] + 3 = x[x]$, जहाँ [x] महत्तम पूर्णांक फलन है : **[JEE(Main)Jan.2023]**

(A) का कोई हल नहीं है (B) का $(-\infty, \infty)$ में ठीक एक हल है

(C) का $(-\infty, 1)$ में ठीक एक हल है (D) का $(-\infty, \infty)$ में ठीक दो हल है

Ans. B

Sol. $x^2 - 4x + [x] + 3 = x[x]$

$\Rightarrow x^2 - 4x + 3 = x[x] - [x]$

$\Rightarrow (x-1)(x-3) = [x] . (x-1)$

$\Rightarrow x = 1$ or $x - 3 = [x]$

$\Rightarrow x - [x] = 3$

$\Rightarrow \{x\} = 3$ (Not Possible)

Only one solution $x = 1$ in $(-\infty, \infty)$

2. Let f(x) be a function such that $f(x + y) = f(x) \cdot f(y)$ for all $x, y \in N$. If $f(1) = 3$ and $\sum_{k=1}^{n} f(k) = 3279$, then the value of n is : **[JEE(Main)Jan.2023]**

माना एक फलन f(x) इस प्रकार है कि सभी $x, y \in N$ के लिए $f(x + y) = f(x) \cdot f(y)$ है। यदि $f(1) = 3$ और $\sum_{k=1}^{n} f(k) = 3279$ है, तब n का मान है : **[JEE(Main)Jan.2023]**

(A) 9 (B) 8 (C) 7 (D) 6

Ans. C

Sol. $f(x + y) = f(x) \cdot f(y) \forall x, y \in N, f(1) = 3$

$f(2) = f^2(1) = 3^2$

$f(3) = f(1)f(2) = 3^3$

$f(4) = 3^4$

$f(k) = 3^k$

$\sum_{k=1}^{n} f(k) = 3279$

$f(1) + f(2) + f(3) + \ldots\ldots\ldots + f(k) = 3279$

$3 + 3^2 + 3^3 + \ldots\ldots\ldots 3^k = 3279$

$$\frac{3\left(3^k-1\right)}{3-1}=3279$$

$$\frac{3^k-1}{2}=1093$$

$$3^k-1=2186$$

$$3^k=2187$$

$$\boxed{k=7}$$

3. If $f(x)=\frac{2^{2x}}{2^{2x}+2}$, $x\in R$, then $f\left(\frac{1}{2023}\right)+f\left(\frac{2}{2023}\right)+.....+f\left(\frac{2022}{2023}\right)$ is equal to :

[JEE(Main)Jan.2023]

यदि $f(x)=\frac{2^{2x}}{2^{2x}+2}$, $x\in R$ है, तब $f\left(\frac{1}{2023}\right)+f\left(\frac{2}{2023}\right)+.....+f\left(\frac{2022}{2023}\right)$ बराबर है :

[JEE(Main)Jan.2023]

(A) 2010 (B) 1010 (C) 2011 (D) 1011

Ans. D

Sol. $f(x)=\frac{4^x}{4^x+2}$

$$f(x)+f(1-x)=\frac{4^x}{4^x+2}+\frac{4^{1-x}}{4^{1-x}+2}$$

$$=\frac{4^x}{4^x+2}+\frac{4}{4+2\left(4^x\right)}$$

$$=\frac{4^x}{4^x+2}+\frac{2}{2+4^x}$$

$= 1$

$\Rightarrow f(x) + f(1-x) = 1$

Now $f\left(\frac{1}{2023}\right)+f\left(\frac{2}{2023}\right)+f\left(\frac{3}{2023}\right)+.....+f\left(1-\frac{3}{2023}\right)+f\left(1-\frac{2}{2023}\right)+f\left(1-\frac{1}{2023}\right)$

Now sum of terms equidistant from beginning and end is 1

Sum = 1 + 1 + 1 + …….+ 1 (1011 times)

= 1011

4. Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by

$f(x)=\log_{\sqrt{m}}\left\{\sqrt{2}(\sin x-\cos x)+m-2\right\}$, for some m, such that the range of f is [0,2]. Then the value of m is _____.

[JEE(Main)Jan.2023]

माना फलन $f:\mathbb{R}\to\mathbb{R}$, किसी m के लिए

$f(x) = \log_{\sqrt{m}}\left\{\sqrt{2}(\sin x - \cos x) + m - 2\right\}$ द्वारा परिभाषित है तथा f का परिसर [0, 2] है। तो m का मान है।

[JEE(Main)Jan.2023]

(A) 3 (B) 2 (C) 5 (D) 4

Ans. C

Sol. $\log_{\sqrt{5}}(\sqrt{2}(\sin x - \cos x) + m - 2) \in [0,2] \in [\log_{\sqrt{5}} 1, \log_{\sqrt{5}} 5]$

$\Rightarrow \sqrt{2}(\sin x - \cos x) + m - 2 \in [1, m]$

Range of $(\sin x - \cos x)$ is $[-\sqrt{2}, \sqrt{2}]$

So, from (i)

$[-4 + m, m] \to [1, m] \Rightarrow -4 + m = 1 \Rightarrow m = 5$

5. Let $f(x) = 2x^n + \lambda, \lambda \in \mathbb{R}, n \in N$, and $f(4) = 133, f(5) = 255$. Then the sum of all the positive integer divisors of $(f(3) - f(2))$ is : **[JEE(Main)Jan.2023]**

माना $f(x) = 2x^n + \lambda, \lambda \in \mathbb{R}, n \in N$, और $f(4) = 133, f(5) = 255$ है। तो $(f(3) - f(2))$ के सभी धनात्मक पूर्णांक भाजकों का योग है : **[JEE(Main)Jan.2023]**

(A) 60 (B) 58 (C) 59 (D) 61

Ans. A

Sol. $f(x) = 2x^n + \lambda$

$f(4) = 133$

$f(5) = 255$

$133 = 2 \times 4^n + \lambda$.... (1)

$255 = 2 \times 5^n + \lambda$.... (2)

(B) – (1)

$122 = 2(5^n - 4^n)$

$\Rightarrow 5^n - 4^n = 61$

$\therefore n = 3$ & $\lambda = 5$

Now, $f(3) - f(2) = 2(3^3 - 2^3) = 38$

divisors = 1,2,19,38

Sum = 60

6. The number of functions $f: \{1,2,3,4\} \to \{a \in \mathbb{Z} : |a| \le 8\}$ satisfying $f(n) + \frac{1}{n} f(n+1) = 1, \forall n \in \{1,2,3\}$ is :

[JEE(Main)Jan.2023]

$f(n) + \frac{1}{n} f(n+1) = 1, \forall n \in \{1,2,3\}$ को संतुष्ट करने वाले फलनों $f: \{1,2,3,4\} \to \{a \in \mathbb{Z} : |a| \le 8\}$ की संख्या है

[JEE(Main)Jan.2023]

(A) 3 (B) 2 (C) 4 (D) 1

Ans. B

Sol. $nf(n) + f(n+1) = n$

$f(1) + f(2) = 1$

$2f(2) + f(3) = 2$

$3f(3) + f(4) = 3$

$2.(1-f(1)) + f(3) = 2$

$f(3) = 2f(1)$

$3.2f(1)+f(4) = 3$

$f(4) = 3 - 6\, f(1) \le 8$

$-8 \le f(4) \le 8$

$-8 \le 3 - 6\, f(1) \le 8$

$-11 \le -6f(1) \le 5$

$$-\frac{5}{6} \le f(1) \le \frac{11}{6}$$

$f(1) = 0, 1$

C–I: $f(1) = 0 , \Rightarrow f(2) = 1$

$f(3) = 0 , f(4) = 3$

C–II: $f(1) = 1 , \Rightarrow f(2) = 0$

$\Rightarrow f(3) = 2 , f(4) = -3$

so '2' such functions.

7. The domain of $f(x) = \dfrac{\log_{(x+1)}(x-2)}{e^{2\log_e x} - (2x+3)}, x \in R$ is : **[JEE(Main)Jan.2023]**

$f(x) = \dfrac{\log_{(x+1)}(x-2)}{e^{2\log_e x} - (2x+3)}, x \in R$ का प्रांत है : **[JEE(Main)Jan.2023]**

(A) $R - \{-1,3\}$ (B) $(-1,\infty) - \{3\}$ (C) $R - \{3\}$ (D) $(2,\infty) - \{3\}$

Ans. D

Sol. $\therefore f(x) = \dfrac{\log_{(x+1)}(x-2)}{e^{2\log x} - (2x+3)}$

(i) $x - 2 > 0$

$x > 2$

$x \in (2, \infty)$

(ii) $x + 1 > 0$ & $x + 1 \ne 1$

$\Rightarrow x > -1$ $\Rightarrow x \ne 0$

$x \in (-1, 0) \cup (0, \infty)$

(iii) $x > 0$

$x \in (0, \infty)$

(iv) $e^{2\ell nx} - (2x+3) \Rightarrow x^2 - 2x - 3 \neq 0$

$\Rightarrow (x-3)(x+1) \neq 0$

$\Rightarrow x \neq 3, x \neq -1$

from (i) $\cap$ (ii) $\cap$ (iii) $\cap$ (iv)

$x \in (2, \infty) - \{3\}$

8. Consider a function $f: \mathbb{N} \to \mathbb{R}$, satisfying

$f(1) + 2f(2) + 3f(3) + ... + xf(x) = x(x+1)f(x);\ x \geq 2$ with $f(1) = 1$.

Then $\frac{1}{f(2022)} + \frac{1}{f(2028)}$ is equal to : **[JEE(Main)Jan.2023]**

एक फलन $f: \mathbb{N} \to \mathbb{R}$, के लिए $f(1) + 2f(2) + 3f(3) + ... + xf(x) = x(x+1)f(x);\ x \geq 2$ तथा $f(1) = 1$ है। तो $\frac{1}{f(2022)} + \frac{1}{f(2028)}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 8000 (B) 8400 (C) 8100 (D) 8200

Ans. C

Sol. f(1) + 2f(2) + 3f(3) …… n f(n) = n (n +1) f(n) ……...(1)

replace in by n + 1

f(1) + 2f(2) + 3f(3) …… nf(n) + (n + 1) f(n+1) = (n + 1) (n +2) f(n +1) ……(2)

Subtracting (1) from (2)

(n +1) f(n +1) = (n +1) (n +2) f(n +1) – n(n +1) f(n) or (n +1)f(n+1) – nf(n) = 0 ……...(3)

put n = 2, 3, 4 …… n + 1 in eq. (3)

3f(3) – 2 f(2) = 0

4f(4) –3f(3) = 0

,,

(n +1) f(n+1) – n f(n) = 0

Add $\Rightarrow$ (n + 1) f(n + 1) = 2f(2)

$\Rightarrow \quad f(n+1) = \frac{2f(2)}{n+1}$ .....(4)

again put n = 2 in eq. (1)

f(1) + 2f(2) = 6f(2)

$\Rightarrow 1 = 4f(2) \Rightarrow f(2) = \frac{1}{4}$

Now by (4) $f(n) = \frac{1}{2n}$

$$\Rightarrow \frac{1}{f(2022)}+\frac{1}{f(2028)}=4044+4056=8100$$

9. The range of the function $f(x)=\sqrt{3-x}+\sqrt{2+x}$ is : **[JEE(Main)Jan.2023]**

फलन $f(x)=\sqrt{3-x}+\sqrt{2+x}$ का परिसर है : **[JEE(Main)Jan.2023]**

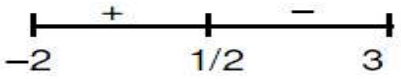
(A) $\left[2\sqrt{2},\sqrt{11}\right]$ (B) $\left[\sqrt{5},\sqrt{13}\right]$ (C) $\left[\sqrt{2},\sqrt{7}\right]$ (D) $\left[\sqrt{5},\sqrt{10}\right]$

Ans. D

Sol. $f'(x)=\frac{1}{2\sqrt{2}+x}-\frac{1}{2\sqrt{3}-x}$

$$\Rightarrow f'(x)=\frac{\sqrt{3-x}-\sqrt{2+x}}{2\sqrt{2+x}\sqrt{3-x}}=0$$

$$\Rightarrow x=\frac{1}{2} \text{ (domain is } -2\le x\le 3)$$



So at $x=\frac{1}{2}$, maxima

$$f(-2)=\sqrt{5}, f\left(\frac{1}{2}\right)=\sqrt{10}, f(3)=\sqrt{5}$$

range $\in \left[\sqrt{5},\sqrt{10}\right]$

10. If the domain of the function $f(x)=\frac{[x]}{1+x^2}$, where [x] is greatest integer $\le x$, is [2,6), then its range is : **[JEE(Main)Jan.2023]**

यदि फलन $f(x)=\frac{[x]}{1+x^2}$ जहाँ [x] महत्तम पूर्णांक $\le x$ है का प्रांत [2,6) है, तो इसका परिसर है **[JEE(Main)Jan.2023]**

(A) $\left(\frac{5}{26},\frac{2}{5}\right]-\left\{\frac{9}{29},\frac{27}{109},\frac{18}{89},\frac{9}{53}\right\}$ (B) $\left(\frac{5}{26},\frac{2}{5}\right]$

(C) $\left(\frac{5}{37},\frac{2}{5}\right]$ (D) $\left(\frac{5}{37},\frac{2}{5}\right]-\left\{\frac{9}{29},\frac{27}{109},\frac{18}{89},\frac{9}{53}\right\}$

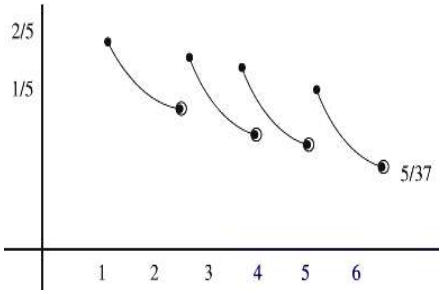
Ans. C

Sol. $f(x)=\frac{2}{1+x^2}$ $x\in[2,3)$

$f(x)=\frac{3}{1+x^2}$ $x\in[3,4)$

$$f(x)=\frac{4}{1+x^2} \qquad x \in [4, 5)$$

$$f(x)=\frac{5}{1+x^2} \qquad x \in [5, 6)$$



$$\left(\frac{5}{37}, \frac{2}{5}\right]$$

11. The absolute minimum value, of the function $f(x) = |x^2 - x + 1| + [x^2 - x + 1]$, where [t] denotes the greatest integer function, in the interval $[-1,2]$, is : **[JEE(Main)Jan.2023]**

अंतराल $[-1, 2]$ में फलन $f(x) = |x^2 - x + 1| + [x^2 - x + 1]$ जहाँ [t] महत्तम पूर्णांक फलन है, का निरपेक्ष न्यूनतम मान है : **[JEE(Main)Jan.2023]**

(A) $\frac{3}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) $\frac{5}{4}$

Ans. B

Sol. $f(x)=\left|x^2-x+1\right|+\left[x^2-x+1\right]; x \in [-1,2]$

Let $g(x) = x^2 - x + 1$

$$=\left(x-\frac{1}{2}\right)^2+\frac{3}{4}$$

$\because \left|x^2-x+1\right|$ and $\left[x^2-x+2\right]$

Both have minimum value at $x=\frac{1}{2}$

$\Rightarrow$ Minimum $f(x) = \frac{3}{4}+0$

$$=\frac{3}{4}$$

12. Let $f : R - \{0,1\} \to R$ be a function such that $f(x)+f\left(\frac{1}{1-x}\right)=1+x$. Then $f(2)$ is equal to : **[JEE(Main)Jan.2023]**

माना फलन $f:R-\{0,1\}\to R$ इस प्रकार है कि $f(x)+f\left(\frac{1}{1-x}\right)=1+x$ है। तो $f(2)$ बराबर है :

[JEE(Main)Jan.2023]

(A) $\frac{9}{4}$ (B) $\frac{7}{4}$ (C) $\frac{9}{2}$ (D) $\frac{7}{3}$

Ans. A

Sol. , $x=2\Rightarrow f(2)+f(-1)=3$ ...(1)

, $x=\frac{1}{2}\Rightarrow f\left(\frac{1}{2}\right)+f(2)=\frac{3}{2}$ ...(2)

, $x=-1\Rightarrow f(-1)+f\left(\frac{1}{2}\right)=0$ ...(3)

(B) ..... (3) $\Rightarrow f(2)-f(-1)=\frac{3}{2}$ ...(4)

$(1)+(4)\Rightarrow 2f(2)=\frac{9}{2}\Rightarrow f(2)=\frac{9}{4}$

13. Let $S=\{1,2,3,4,5,6\}$. Then the number of one-one function $f: S\to P(S)$, where P(S) denote the power set of S, such that $f(n)\subset f(m)$ where $n<m$ is _____ . **[JEE(Main)Jan.2023]**

माना $S=\{1,2,3,4,5,6\}$ है तो ऐसे एकैकी फलनों $f: S\to P(S)$, जहाँ P(S) समुच्चय S का घात समुच्चय $f(n)\subset f(m)$ है जब कि $n<m$ है, की संख्या है। **[JEE(Main)Jan.2023]**

Ans. 3240

Sol. Let $S=\{1, 2, 3, 4, 5, 6\}$, then the number of one-one functions, $f: S\bullet P(S)$, where P(S) denotes the power set of S, such that $f(n)<f(m)$ where $n<m$ is

$n(S)=6$

$P(S)=\{\phi,\{1\},.....\{6\},\{1,2\},....,\{5.6\},.....,\{1,2,3,4,5,6,\}\}$

–64 elements

case – 1

$f(6)$ = S i.e. 1 option,

$f(5)$ = any 5 element subset A of S i.e. 6 options,

$f(4)$ = any 4 element subset B of A i.e. 5 options,

$f(3)$ = any 3 element subset C of B i.e. 4 options,

$f(2)$ = any 2 element subset D of C i.e. 3 options,

$f(1)$ = any 1 element subset E of D or empty subset i.e. 3 options,

Total functions = 1080

Case – 2

$f(6)$ = any 5 element subset A of S i.e. 6 options,

$f(5)$ = any 4 element subset B of A i.e. 5 options,

f? (4) = any 3 element subset C of B i.e. 4 options,

$f(3)$ = any 2 element subset D of C i.e. 3 options,

f? (2) = any 1 element subset E of D i.e. 2 options,
f (1) = empty subset i.e. 1 option
Total functions = 720
Case – 3
f(6)= S
f(5) = any 4 element subset A of ' S i.e. 15 options,
f(4) = any 3 element subset B of A i.e. 4 options,
f(3) = any 2 element subset C of B i.e. 3 options,
f(2) = any 1 element subset D of C i.e. 2 options,
f(1) = empty subset i.e. 1 option
Total functions = 360
Case – 4
f(6) = S
f(5) = any 5 element subset A of S i.e. 6 options,
f(4) = any 3 element subset B of A i.e. 10 options,
f(3) = any 2 element subset C of B i.e. 3 options,
f(2) = any 1 element subset D of C i.e. 2 options,
f(1) = empty subset i.e. 1 option
Total functions = 360
Case – 5
f(6) = S
f(5) = any 5 element subset A of S i.e. 6 options,
f(4) = any 4 element subset B of A i.e. 5 options,
f(3) = any 2 element subset C of B i.e. 6 options,
f(2) = any 1 element subset D of C i.e. 2 options,
f(1) = empty subset i.e. 1 option
Total functions = 360
Case – 6
f(6) = S
f(5) = any 5 element subset A of S i.e. 6 options,
f(4) = any 4 element subset B of A i.e. 5 options,
f(3) = any 3 element subset C of B i.e. 4 options,
f(2) = any 1 element subset D of C i.e. 3 options,
f(1) = empty subset i.e. 1 option
Total functions = 360
$\therefore$ Number of such functions = 3240

14. For some a, b, c $\in$ N, let f(x) = ax – 3 and g(x) = x^b + c, x $\in$ R. If $\left(fog\right)^{-1}\left(x\right)=\left(\frac{x-7}{2}\right)^{\frac{1}{3}}$, then (fog) (ac) + (gof) (b) is equal to _____. **[JEE(Main)Jan.2023]**

किसी a, b, c $\in$ N के लिए, माना f(x) = ax – 3 तथा g(x) = x^b + c, x $\in$ R है। यदि $\left(fog\right)^{-1}\left(x\right)=\left(\frac{x-7}{2}\right)^{\frac{1}{3}}$ है, तो (fog) (ac) + (gof) (b) बराबर है। **[JEE(Main)Jan.2023]**

Ans. 2039

Sol. Let $fog(x) = h(x)$

$\Rightarrow h^{-1}(x) = \left(\frac{x-7}{2}\right)^{\frac{1}{3}}$

$\Rightarrow h(x) = fog(x) = 2x^3 + 7$

$fog(x) = a(xb + c) - 3$

$\Rightarrow a = 2, b = 3, c = 5$

$\Rightarrow fog(ac) = fog(10) = 2007$

$g(f(x) = (2x - 3)^3 + 5$

$\Rightarrow gof(b) = gof(3) = 32$

$\Rightarrow sum = 2039$

15. Suppose f is a function satisfying $f(x+y) = f(x) + f(y)$ for all, $x, y \in \mathbb{N}$ and $f(1) = \frac{1}{5}$. If $\sum_{n=1}^{m} \frac{f(n)}{n(n+1)(n+2)} = \frac{1}{12}$, then m is equal to _____. **[JEE(Main)Jan.2023]**

माना f एक फलन है जो सभी $x, y \in \mathbb{N}$ के लिए $f(x + y) = f(x) + f(y)$ को संतुष्ट करता है एवं $f(1) = \frac{1}{5}$ है। यदि $\sum_{n=1}^{m} \frac{f(n)}{n(n+1)(n+2)} = \frac{1}{12}$ है, तब m बराबर है। **[JEE(Main)Jan.2023]**

Ans. 10

Sol. $\because f(1) = \frac{1}{5} \therefore f(2) = f(1) + f(1) = \frac{2}{5}$

$f(2) = \frac{2}{5}$ $f(3) = f(2) + f(1) = \frac{3}{5}$

$f(3) = \frac{3}{5}$

$\therefore \sum_{n=1}^{m} \frac{f(n)}{n(n+1)(n+2)}$

$= \frac{1}{5} \sum_{n=1}^{m} \left(\frac{1}{n+1} - \frac{1}{n+2}\right)$

$= \frac{1}{5}\left(\frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + ... + \frac{1}{m+1} - \frac{1}{m+2}\right)$

$= \frac{1}{5}\left(\frac{1}{2} - \frac{1}{m+2}\right) = \frac{m}{10(m+2)} = \frac{1}{12}$

$\therefore m = 10$

16. Let $f^1(x) = \frac{3x+2}{2x+3}$, $x \in R - \left\{\frac{-3}{2}\right\}$; For $n \geq 2$, define $f^n(x) = f^1 o f^{n-1}(x)$. If $f^5(x) = \frac{ax+b}{bx+a}$, gcd(a,b) = 1, then

a + b is equal to _____ . **[JEE(Main)Jan.2023]**

माना $f^1(x)=\frac{3x+2}{2x+3}, x \in R-\left\{\frac{-3}{2}\right\}$ है। $n \geq 2$, के लिए $f^n(x)=f^1 o f^{n-1}(x)$ द्वारा परिभाषित कीजिए। यदि $f^5(x)=\frac{ax+b}{bx+a}$, gcd(a,b) = 1 है, तो a + b बराबर है। **[JEE(Main)Jan.2023]**

Ans. 3125

Sol. $f^1(x)=\frac{3x+2}{2x+3}$

$\Rightarrow f^2(x)=\frac{13x+12}{12x+13}$

$\Rightarrow f^3(x)=\frac{63x+62}{62x+63}$

$\therefore f^5(x)=\frac{1563x+1562}{1562x+1563}$

a + b = 3125

17. Let A= {1,2,3,5,8,9}. Then the number of possible functions $f: A \to A$ such that $f(m \cdot n)=f(m) \cdot f(n)$ for every $m, n \in A$ with $m \cdot n \in A$ is equal to _____ . **[JEE(Main)Jan.2023]**

माना A= {1,2,3,5,8,9} है। तब संभव फलनों $f: A \to A$, जब कि $m, n \in A$ तथा $m \cdot n \in A$ को संतुष्ट करने वाले सभी m, n के लिए $f(m \cdot n)=f(m) \cdot f(n)$ है, की संख्या है। **[JEE(Main)Jan.2023]**

Ans. 432

Sol. f(1) = 1; f(9) = f(3) × f(3)

i.e., f(3) = 1 or 3

Total function = 1 × 6 × 2 × 6 × 6 × 1 = 432

18. Let $A=\{x \in \mathbb{R}:[x+3]+[x+4] \leq 3\}$, $B=\left\{x \in \mathbb{R}: 3^x\left(\sum_{r=1}^{\infty}\frac{3}{10^r}\right)^{x-3}<3^{-3x}\right\}$, where [t] denotes greatest integer function. Then : **[JEE(Main)April,2023]**

माना $A=\{x \in \mathbb{R}:[x+3]+[x+4] \leq 3\}$, $B=\left\{x \in \mathbb{R}: 3^x\left(\sum_{r=1}^{\infty}\frac{3}{10^r}\right)^{x-3}<3^{-3x}\right\}$ है, जहाँ [t] महत्तम पूर्णांक फलन है। तब: **[JEE(Main)April,2023]**

(A) A = B (B) $B \subset C, A \neq B$ (C) $A \cap B=\phi$ (D) $A \subset B, \quad A \neq B$

Ans. A

Sol. $[x]+3+[x]+4 \leq 3$

$2[x] \leq -4$

$[x] \leq -2 \Rightarrow x \in (-\infty,-1)$. _____A

$$3^x\left(\frac{3\cdot\frac{1}{10}}{1-\frac{1}{10}}\right)^{x-3} < 3^{-3x}$$

$27 < 3^{-3x}$

$-3x > +3$

$x < -1$ _____B

$A = B$

19. Let the sets A and B denote the domain and range respectively of the function $f(x)=\frac{1}{\sqrt{[x]-x}}$, where [x] denotes the smallest integer greater than or equal to x. Then among the statements :

(S1) : $A \cap B = (1,\infty) - \mathbb{N}$ and

(S2) : $A \cup B = (1,\infty)$ **[JEE(Main)April,2023]**

(A) only (S2) is true (B) neither (S1) nor (S2) is true

(C) only (S1) is true (D) both (S1) and (S2) are true

माना फलन $f(x)=\frac{1}{\sqrt{[x]-x}}$, जहाँ [x] न्यूनतम पूर्णांक $\geq x$ है, के प्रांत तथा परिसर क्रमशः समुच्चय A तथा B है। तो कथनों

(S1) : $A \cap B = (1,\infty) - \mathbb{N}$ तथा

(S2) : $A \cup B = (1,\infty)$ **[JEE(Main)April,2023]**

(A) केवल (S2) सही है (B) न तो (S1) न ही (S2) सही है

(C) केवल (S1) सही है (D) दोनों (S1) तथा (S2) सही है

Ans. C

Sol. $[x] - x > 0 \Rightarrow [x] > x$

$\Rightarrow x \notin I$ So $A = R - I$

Also $0 < [x] - x < 1$ when $x \in I$

So $f(x) \in (1,\infty) \Rightarrow B = (1,\infty)$

$A \cap B = (1,\infty) - N$

$A \cup B = (1,\infty) \cup \{R - I\}$

So S_1 is true & S_2 is false

20. If $f(x)=\frac{\left(\tan 1^\circ\right)x + \log_e(123)}{x\log_e(1234) - \left(\tan 1^\circ\right)}, x > 0$, then the least value of $f(f(x)) + f\left(f\left(\frac{4}{x}\right)\right)$ is :

[JEE(Main)April,2023]

यदि $f(x)=\frac{\left(\tan 1^{\circ}\right)x+\log_e(123)}{x\log_e(1234)-\left(\tan 1^{\circ}\right)}, x>0$ है, तो $f(f(x))+f\left(f\left(\frac{4}{x}\right)\right)$ का निम्नतम मान है :

[JEE(Main)April,2023]

(A) 2 (B) 8 (C) 4 (D) 0

Ans. C

Sol. $f(x)=\frac{(\tan 1)x+\log 123}{x\log 1234-\tan 1}$

Let $A=\tan 1, B=\log 123, C=\log 1234$

$f(x)=\frac{Ax+B}{xC-A}$

$f(f(x))=\frac{A\left(\frac{Ax+B}{xC-A}\right)+B}{C\left(\frac{Ax+B}{CX-A}\right)-A}$

$=\frac{A^2x+AB+xBC-AB}{ACx+BC-ACx+A^2}$

$=\frac{x\left(A^2+BC\right)}{\left(A^2+BC\right)}=x$

$f(f(x))=x$

$f\left(f\left(\frac{4}{x}\right)\right)=\frac{4}{x}$

$f(f(x))+f\left(f\left(\frac{4}{x}\right)\right)$

$AM \geq GM$

$x+\frac{4}{x}\geq 4$

21. The domain of the function $f(x)=\frac{1}{\sqrt{[x]^2-3[x]-10}}$ is : (where [x] denotes the greatest integer less than or equal to x) :

[JEE(Main)April,2023]

फलन $f(x)=\frac{1}{\sqrt{[x]^2-3[x]-10}}$, जहाँ [x] महत्तम पूर्णांक $\leq x$ है, का प्रांत है :

[JEE(Main)April,2023]

(A) $(-\infty,-3]\bigcup(5,\infty)$ (B) $(-\infty,-2)\cup[6,\infty)$ (C) $(-\infty,-2)\cup(5,\infty)$ (D) $(-\infty,-3]\cup[6,\infty)$

Ans. B

Sol. $f(x)=\frac{1}{\sqrt{[x]^2-3[x]-10}}$

For Domain $[x]^2-3[x]-10>0$

$\Rightarrow \quad ([x]-5)([x]+2)>0$

$\Rightarrow \quad [x]\in(-\infty,-2)\cup(5,\infty)$

$\therefore \quad x\in(-\infty,-2)\cup[6,\infty)$

22. For $x\in\mathbb{R}$, two real valued functions f(x) and g(x) are such that, $g(x)=\sqrt{x}+1$ and $fog(x)=x+3-\sqrt{x}$. Then f(0) is equal to : **[JEE(Main)April,2023]**

$x\in\mathbb{R}$ के लिए दो वास्तविक फलन f(x) तथा g(x) इस प्रकार है कि $g(x)=\sqrt{x}+1$ तथा $fog(x)=x+3-\sqrt{x}$ हैं। तो f(0) बराबर है : **[JEE(Main)April,2023]**

(A) 5 (B) 0 (C) –3 (D) 1

Ans. (Bonus)

Sol. $g(x)=\sqrt{x}+1$

$fog(x)=x+3-\sqrt{x}$

$=\left(\sqrt{x}+1\right)^2-3\left(\sqrt{x}+1\right)+5$

$= g^2(x) - 3g(x) + 5$

$\Rightarrow f(x) = x^2 - 3x + 5$

$\therefore f(0) = 5$

But, if we consider the domain of the composite function fog(x) then in that case f(0) will not be defined as g(x) cannot be equal to zero.

23. Let R = {a, b, c, d, e} and S = {1, 2, 3, 4}. Total number of onto functions f: R → S such that f(a) ≠ 1, is equal to _____. **[JEE(Main)April,2023]**

माना R = {a, b, c, d, e} तथा S = {1, 2, 3, 4} हैं। आच्छादक फलनों f : R → S जिनके लिए f(a) ≠ 1 है, की कुल संख्या है : **[JEE(Main)April,2023]**

Ans. 384

Ans. Matrix 180

Sol. Total onto function

$\frac{\lfloor 5}{\lfloor 3\lfloor 2}\times\lfloor 4=240$

Now when f(a) = 1

$\lfloor 4+\frac{\lfloor 4}{\lfloor 2\lfloor 2}\times\lfloor 3=24+36=60$

so required $f^n = 240 - 60 = 180$

24. Let m and n be the numbers of real roots of the quadratic equations $x^2 - 12x + [x] + 31 = 0$ and $x^2 - 5|x + 2| - 4 = 0$ respectively, where [x] denotes the greatest integer $\le x$. Then $m^2 + mn + n^2$ is equal to _____. **[JEE(Main)April,2023]**

माना समीकरणों $x^2 - 12x + [x] + 31 = 0$ तथा $x^2 - 5|x + 2| - 4 = 0$ के वास्तविक मूलों की संख्या क्रमशः m तथा n है, जहाँ [x] महत्तम पूर्णांक $\le x$ है। तो $m^2 + mn + n^2$ बराबर है : **[JEE(Main)April,2023]**

Ans. 9

Sol. $x^2 - 12x + [x] + 31 = 0$

$\underbrace{x^2 - 12x + 31}_{\ge -5} + [x] = 0$

It could have its solution in [5, 6] but it does not exist as at x = 5 and 6

LHS = 1

m = 0

$x^2 - 5|x + 2| - 4 = 0$

Case 1

$x \ge -2$

$x^2 - 5(x + 2) - 4 = 0$

$x^2 - 5x - 14 = 0 \Rightarrow x = 7, -2$

Case 2

$x < -2$

$x^2 + 5x + 10 - 4 = 0$

$x = -2, -3$

3 solution i.e., $x = -3, -2, 7$

$n = 3$

25. Let A = {1, 2, 3, 4, 5} and B = {1, 2, 3, 4, 5, 6}. Then the number of functions $f : A \to B$ satisfying $f(1) + f(2) = f(4) - 1$ is equal to _____. **[JEE(Main)April,2023]**

माना A = {1, 2, 3, 4, 5} तथा B = {1, 2, 3, 4, 5, 6} हैं। तो $f(1) + f(2) = f(4) - 1$ को संतुष्ट करने वाले फलनों $f : A \to B$ की संख्या है _____ **[JEE(Main)April,2023]**

Ans. 360

Sol. $f(1) + f(2) + 1 = f(4)$

$\Rightarrow f(1) + f(2) \le 5$

Let $f(4) = 6$, then $f(1) + f(2) = 5$

So (f_1, f_2) may be (1, 4), (4, 1), (2, 3), (3, 2)

Total ways = 4 . 1 . 6 . 6 = 144

Let f(4) = 5, $f(1) + f(2) = 4$

so (f_1, f_2) may be (1, 3), (3, 1), (2, 2)

Total ways = 3 . 1 . 6 . 6 = 108

Let f(4) = 4, $f(1) + f(2) = 3$

so (f_1, f_2) may be (1, 2), (2, 1)

Total ways = 2 . 1 . 6 . 6 = 72

Let f(4) = 3, f(1) + f(2) = 2

Only way (1, 1)

Total ways = 1 . 1 . 36 = 36

26. Let $[\alpha]$ denote the greatest integer $\leq \alpha$. Then $\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+\ldots+\left[\sqrt{120}\right]$ is equal to _____.

[JEE(Main)April,2023]

माना $[\alpha]$ महत्तम पूर्णांक $\leq \alpha$ है। तो $\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+\ldots+\left[\sqrt{120}\right]$ बराबर है _____

[JEE(Main)April,2023]

Ans. 825

Sol. $S=\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+....+\left[\sqrt{120}\right]$

$\left[\sqrt{1}\right]\rightarrow\left[\sqrt{3}\right]=1\times3$

$\left[\sqrt{4}\right]\rightarrow\left[\sqrt{8}\right]=2\times5$

$\left[\sqrt{9}\right]\rightarrow\left[\sqrt{15}\right]=3\times7$

$\vdots$

$\left[\sqrt{100}\right]\rightarrow\left[\sqrt{120}\right]=10\times21$

$S=1\times3+2\times5+3\times7+....+10\times21$

$=\sum_{r=1}^{10} r(2r+1)$

$=2\sum_{r=1}^{10} r^2+\sum_{r=1}^{10} r$

$=\frac{2\times10\times11\times21}{6}+\frac{10\times11}{2}$

$= 770 + 55$

$= 825$

27. Let $f(x)=\sum_{k=1}^{10} k\,x^k, x\in\mathbb{R}$. If $2f(2)+f'(2)=119(2)^n+1$ then n is equal to _____.

[JEE(Main)April,2023]

माना $f(x)=\sum_{k=1}^{10} k\,x^k, x\in\mathbb{R}$ है। यदि $2f(2)+f'(2)=119(2)^n+1$ है, तो n बराबर है _____

[JEE(Main)April,2023]

Ans. 10

Sol. $f(x)=\sum_{k=1}^{10} kx^k$

$f(x) = x + 2x^2 + ... + 10\,x^{10}$

$f(x) \cdot x = x^2 + 2x^3 + ... + 9x^{10} + 10x^{11}$

$f(x)\,(1 - x) = x + x^2 + x^3 + ... + x^{10} - 10\,x^{11}$

$$f(x)=\frac{x\left(1-x^{10}\right)}{(1-x)^2}-\frac{10x^{11}}{(1-x)}$$

$$f(x)=\frac{x-x^{11}-10x^{11}+10x^{12}}{(1-x)^2} \Rightarrow \frac{10x^{12}-11x^{11}+x}{(1-x)^2}$$

Hence $2f(2) + f'(2) = 119.2^{10} + 1$

$\Rightarrow$ So, n = 10

28. The function $f : N - \{1\} \to N$; is defined by f(n) = the highest prime factor of n, is :

[JEE(Main)Jan.2024]

(A) neither one-one nor onto (B) onto only

(C) one-one only (D) both one-one and onto

फलन $f : N - \{1\} \to N$; f(n) = पूर्णांक n का सबसे बड़ा अभाज्य गुणनखंड, द्वारा परिभाषित है। तो f :

[JEE(Main)Jan.2024]

(A) न तो एकैकी है न ही आच्छादक है (B) केवल आच्छादक है

(C) केवल एकैकी है (D) एकैकी और आच्छादक दोनों है

Ans. A

Sol. Largest prime factors of two different natural numbers can be same so it is many one.

For example f(6) = 3, f(12) = 3

f(10) = 5, f(15) = 5

As range of f(x) is set of prime number so range of f(x) is proper subset of co-domain so f is into.

29. Let $f : R - \left\{\frac{-1}{2}\right\} \to R$ and $g : R - \left\{\frac{-5}{2}\right\} \to R$ be defined as $f(x)=\frac{2x+3}{2x+1}$ and $g(x)=\frac{|x|+1}{2x+5}$. Then, the domain of the function fog is : **[JEE(Main)Jan.2024]**

माना $f : R - \left\{\frac{-1}{2}\right\} \to R$ तथा $g : R - \left\{\frac{-5}{2}\right\} \to R$, $f(x)=\frac{2x+3}{2x+1}$ तथा $g(x)=\frac{|x|+1}{2x+5}$ द्वारा परिभाषित हैं। तो फलन fog का प्रांत है : **[JEE(Main)Jan.2024]**

(A) $R - \left\{-\frac{7}{4}\right\}$ (B) R (C) $R - \left\{-\frac{5}{2}\right\}$ (D) $R - \left\{-\frac{5}{2}, -\frac{7}{4}\right\}$

Ans. C

Sol. $$y=\frac{\frac{2(|x|+1)}{2x+5}+3}{2\left(\frac{|x|+1}{2x+5}\right)+1}$$

$$=\frac{2|x|+2+6x+15}{2|x|+2+2x+5}=\frac{(2|x|+6x+17)}{2|x|+2x+7}$$

And $x \neq -\frac{5}{2}$ for defining g(x)

30. If $f(x)=\begin{cases}2+2x, & -1\le x<0\\ 1-\frac{x}{3}, & 0\le x\le 3\end{cases}$; $g(x)=\begin{cases}-x, & -3\le x\le 0\\ x, & 0<x\le 1\end{cases}$, then range of (fog) (x) is :

[JEE(Main)Jan.2024]

यदि $f(x)=\begin{cases}2+2x, & -1\le x<0\\ 1-\frac{x}{3}, & 0\le x\le 3\end{cases}$; $g(x)=\begin{cases}-x, & -3\le x\le 0\\ x, & 0<x\le 1\end{cases}$, तो (fog) (x) का परिसर है :

[JEE(Main)Jan.2024]

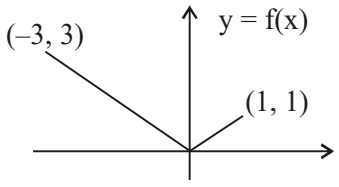
(A) [0, 1) (B) [0, 3) (C) (0, 1] (D) [0, 1]

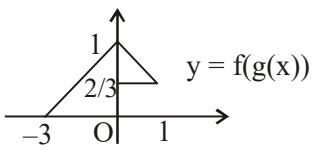
Ans. D

Sol. $$f(g(x))=\begin{cases}2+2g(x) & , \quad -3\le g(x)<0 \quad ____(1)\\ 1-\frac{g(x)}{3} & , \quad 0\le g(x)\le 3 \quad ____(2)\end{cases}$$

By (1) $x \in \phi$

And by (2) $x \in [-3, 0]$ and $x \in [0, 1]$





Range of f(g(x)) is [0, 1]

31. If the domain of the function $f(x)=\log_e\left(\frac{2x+3}{4x^2+x-3}\right)+\cos^{-1}\left(\frac{2x-1}{x+2}\right)$ is $(\alpha, \beta]$ then the value of $5\beta-4\alpha$ is equal to : **[JEE(Main)Jan.2024]**

यदि फलन $f(x)=\log_e\left(\frac{2x+3}{4x^2+x-3}\right)+\cos^{-1}\left(\frac{2x-1}{x+2}\right)$ का प्रांत $(\alpha, \beta]$ है, तो $5\beta - 4\alpha$ का मान बराबर है :

[JEE(Main)Jan.2024]

(A) 10 (B) 9 (C) 12 (D) 11

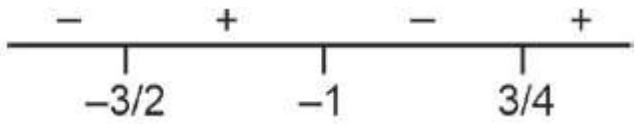
Ans. C

Sol. Given function,

$$f(x)=\log_e\left(\frac{2x+3}{4x^2+x-3}\right)+\cos^{-1}\left(\frac{2x-1}{x+2}\right)$$

Clearly, $\frac{2x+3}{4x^2+x-3}>0$

$$\Rightarrow \frac{2x+3}{(4x-3)(x+1)}>0$$

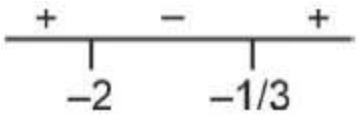


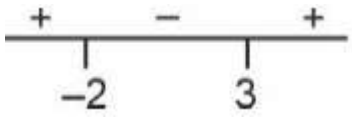
$$\therefore x\in\left(\frac{-3}{2},-1\right)\cup\left(\frac{3}{4},\infty\right)$$

Also, $-1\le\frac{2x-1}{x+2}\le 1$

$$\Rightarrow \frac{2x-1}{x+2}+1\ge 0 \qquad \frac{2x-1}{x+2}-1\le 0$$

$$\Rightarrow \frac{3x+1}{x+2}\ge 0 \qquad \frac{x-3}{x+2}\le 0$$





$$\therefore x\in(-\infty,-2)\cup\left[-\frac{1}{3},\infty\right) \text{ and } x\in(-2,3]$$

$$\therefore x\in\left(\frac{3}{4},3\right]\equiv(\alpha,\beta]$$

$\therefore 5\beta - 4\alpha = 15 - 3 = 12$

32. If $f(x)=\frac{4x+3}{6x-4}$, $x\ne\frac{2}{3}$ and $(fof)(x)=g(x)$, where $g:R-\left\{\frac{2}{3}\right\}\to R-\left\{\frac{2}{3}\right\}$, then $(gogog)(4)$ is equal to :

[JEE(Main)Jan.2024]

यदि $f(x)=\frac{4x+3}{6x-4}$, $x \neq \frac{2}{3}$ तथा (fof) (x) = g(x) है, जहाँ $g: R-\left\{\frac{2}{3}\right\} \to R-\left\{\frac{2}{3}\right\}$ है, तो (gogog) (4) बराबर है:

[JEE(Main)Jan.2024]

(A) –4 (B) $-\frac{19}{20}$ (C) $\frac{19}{20}$ (D) 4

Ans. D

Sol. $g(x)=f\left(\frac{4x+3}{6x-4}\right)=\frac{4\left(\frac{4x+3}{6x-4}\right)+3}{6\left(\frac{4x+3}{6x-4}\right)-4}=\frac{16x+12+18x-12}{24x+18-24x+16}=\frac{34x}{34}=x$

$gogog(4)=g(g(4))=g(4)=4$

33. Let $f: R \to R$ and $g: R \to R$ be defined as $f(x)=\begin{cases}\log_e x & , \quad x>0 \\ e^{-x} & , \quad x \leq 0\end{cases}$ and $g(x)=\begin{cases}x & , \quad x \geq 0 \\ e^{x} & , \quad x<0\end{cases}$. Then $gof: R \to R$ is:

[JEE(Main)Jan.2024]

(A) both one-one and onto (B) onto but not one-one

(C) one-one but not onto (D) neither one-one nor onto

माना $f: R \to R$ तथा $g: R \to R$, $f(x)=\begin{cases}\log_e x & , \quad x>0 \\ e^{-x} & , \quad x \leq 0\end{cases}$ तथा $g(x)=\begin{cases}x & , \quad x \geq 0 \\ e^{x} & , \quad x<0\end{cases}$ द्वारा परिभाषित है। तो $gof: R \to R$:

[JEE(Main)Jan.2024]

(A) एकैकी एवं आच्छादक दोनों है (B) आच्छादक है परन्तु एकैकी नहीं है

(C) एकैकी है परन्तु आच्छादक नहीं है (D) न तो एकैकी है, न ही आच्छादक है

Ans. D

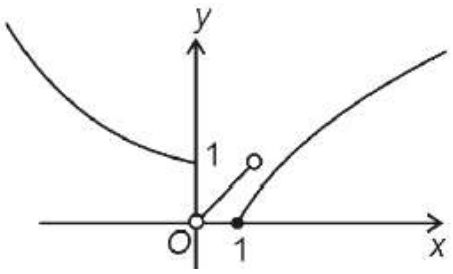
Sol. Given $f(x)=\begin{cases}e^{-x} & ; \quad x \leq 0 \\ \ell nx & ; \quad x>0\end{cases}$

and $g(x)=\begin{cases}x & ; \quad x \geq 0 \\ e^{x} & ; \quad x<0\end{cases}$

then gof(x)
= g(f(x))

$=\begin{cases}f(x) & ; \quad f(x) \geq 0 \\ e^{f(x)} & ; \quad f(x)<0\end{cases}$

$$= \begin{cases} e^{-x} & ; \quad x \le 0 \\ x & ; \quad 0 < x < 1 \\ \ell nx & ; \quad x \ge 1 \end{cases}$$



gof(x)

$\therefore$ f(x) is neither one-one nor onto

34. If the domain of the function $f(x) = \frac{\sqrt{x^2-25}}{(4-x^2)} + \log_{10}(x^2+2x-15)$ is $(-\infty,\alpha)\cup[\beta,\infty)$, then $\alpha^2+\beta^3$ is equal to : **[JEE(Main)Jan.2024]**

यदि फलन $f(x) = \frac{\sqrt{x^2-25}}{(4-x^2)} + \log_{10}(x^2+2x-15)$ का प्राँत $(-\infty,\alpha)\cup[\beta,\infty)$ है, तो $\alpha^2+\beta^3$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 175 (B) 140 (C) 150 (D) 125

Ans. C

Sol. $x^2-25 \ge 0 \quad \Rightarrow x \in (-\infty,-5]\cup[5,\infty)$

$4-x^2 \ne 0 \quad \Rightarrow x \notin \{-2,2\}$

$x^2+2x-15 > 0 \quad \Rightarrow x \in (-\infty,-5)\cup(3,\infty)$

$\Rightarrow x \in (-\infty,-5)\cup[5,\infty)$

$\Rightarrow \alpha = -5, \beta = 5$

35. Let A = {1, 2, 3, ..., 7} and let P(A) denote the power set of A. If the number of functions $f: A \to P(A)$ such that $a \in f(a)$, $\forall a \in A$, is m^n, m and $n \in N$ and m is least, then m + n is equal to _____. **[JEE(Main)Jan.2024]**

माना A = {1, 2, 3, ..., 7} है तथा माना समुच्चय A का घात (Power) समुच्चय P(A) है। यदि फलनों $f: A \to P(A)$, जिनके लिए $a \in f(a)$, $\forall a \in A$ है, की संख्या m^n, m, $n \in N$ हैं तथा m न्यूनतम है, तो m + n बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. 44

Sol. $n(P(A)) = 2^7$

$a \in f(a)$

$\Rightarrow$ f(a) will have (2^6) different subsets having a in them as choice.

$\Rightarrow (2^6)^7 = 2^{42}$ function

$m^n = 2^{42}$

$\Rightarrow m + n = 44$

36. Let A = {1, 3, 7, 9, 11} and B = {2, 4, 5, 7, 8, 10, 12}. Then the total number of one-one maps f: A $\rightarrow$ B, such that f(1) + f(3) = 14, is : **[JEE(Main)April,2024]**

माना A = {1, 3, 7, 9, 11} तथा B = {2, 4, 5, 7, 8, 10, 12} हैं। तो एकैकी प्रतिचित्रों (maps) f: A $\rightarrow$ B, जिनके लिए f(1) + f(3) = 14 है, की कुल संख्या है : **[JEE(Main)April,2024]**

(A) 180 (B) 120 (C) 480 (D) 240

Ans. D

Sol. A = {1, 3, 7, 9, 11}

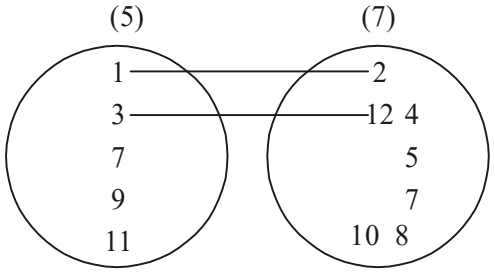
B = {2, 4, 5, 7, 8, 10, 12}

f(1) + f(3) = 14

(i) 2 + 12

(ii) 4 + 10

$2 \times (2 \times 5 \times 4 \times 3) = 240$



37. Let f, g : R $\rightarrow$ R be defined as :

$$f(x) = |x-1| \text{ and } g(x) = \begin{cases} e^x, & x \geq 0 \\ x+1, & x \leq 0 \end{cases}$$

Then the function f(g(x)) is : **[JEE(Main)April,2024]**

(A) Neither one-one nor onto (B) Onto but not one-one

(C) One-one but not onto (D) Both one-one and onto

माना f, g : R $\rightarrow$ R,

$$f(x) = |x-1| \text{ तथा } g(x) = \begin{cases} e^x, & x \geq 0 \\ x+1, & x \leq 0 \end{cases}$$

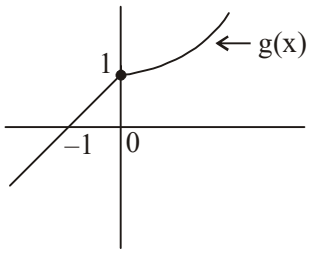
द्वारा परिभाषित हैं। तो फलन f(g(x)) : **[JEE(Main)April,2024]**

(A) न तो एकैकी न ही आच्छादक है (B) आच्छादक है परन्तु एकैकी नहीं है

(C) एकैकी है परन्तु आच्छादक नहीं है (D) एकैकी तथा आच्छादक दोनों है

Ans. A

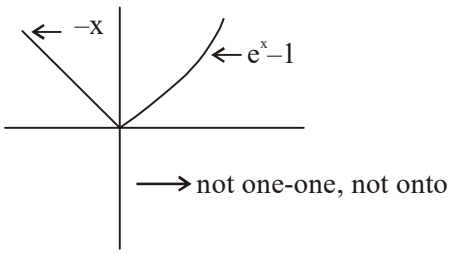
Sol. $f(x)=\begin{cases}x-1 & x\geq 1\\ 1-x & x<1\end{cases}$



$$g(x)=\begin{cases}e^x & ; \quad x\geq 0\\ x+1 & ; \quad x\leq 0\end{cases}$$

$$f(g(x))=\begin{cases}g(x)-1, & g(x)\geq 1\\ 1-g(x) & g(x)<1\end{cases}$$

$$=\begin{cases}e^x-1 & x\geq 0\\ -x & x<0\end{cases}$$



38. The function $f(x)=\dfrac{x^2+2x-15}{x^2-4x+9}, x\in\mathbb{R}$ is: **[JEE(Main)April,2024]**

(A) both one-one and onto (B) onto but not one-one

(C) neither one-one nor onto (D) one-one but not onto

फलन $f(x)=\dfrac{x^2+2x-15}{x^2-4x+9}, x\in\mathbb{R}$ है : **[JEE(Main)April,2024]**

(A) एकैकी तथा आच्छादक दोनों (B) आच्छादक किंतु एकैकी नहीं

(C) न तो एकैकी और न आच्छादक (D) एकैकी किंतु आच्छादक नहीं

Ans. C

Sol. $f'(x)=\dfrac{\left(x^2-4x+9\right)(2x+2)-\left(x^2+2x-15\right)(2x-4)}{\left(x^2-4x+9\right)^2}$

$=\dfrac{2x^3-6x^2+10x+18-\left[2x^3-38x+60\right]}{\left(x^2-4x+9\right)^2}$

$=\dfrac{-6x^2+48x-42}{\left(x^2-4x+19\right)^2}=\dfrac{-6(x-7)(x-1)}{\left(x^2-4x+9\right)^2}$

So f'(x) = positive as well as negative so many one and into.

Since codomain is not given so will be considered it R

39. Let $f(x)=\dfrac{1}{7-\sin 5x}$ be a function defined on R. Then the range of the function f(x) is equal to :

[JEE(Main)April,2024]

माना फलन $f(x)=\dfrac{1}{7-\sin 5x}$, R पर परिभाषित है। तो फलन f(x) का परिसर बराबर है : **[JEE(Main)April,2024]**

(A) $\left[\dfrac{1}{8},\dfrac{1}{6}\right]$ (B) $\left[\dfrac{1}{7},\dfrac{1}{5}\right]$ (C) $\left[\dfrac{1}{8},\dfrac{1}{5}\right]$ (D) $\left[\dfrac{1}{7},\dfrac{1}{6}\right]$

Ans. A

Sol. $f(x)=\dfrac{1}{7-\sin x}, x\in R$

$-1 \le -\sin 5x \le 1$

$-1 \le -\sin -5x \le 1$

$6 \le 7-\sin 5x \le 8$

$\dfrac{1}{8} \le \dfrac{1}{7-\sin 5x} \le \dfrac{1}{6}$

$\dfrac{1}{8} \le f(x) \le \dfrac{1}{6}$

40. Let [t] be the greatest integer less than or equal to t. Let A be the set of all prime factors of 2310 and $f: A \to Z$ be the function $f(x)=\left[\log_2\left(x^2+\left[\dfrac{x^3}{5}\right]\right)\right]$. The number of one-to-one functions from A to the range of f is :

[JEE(Main)April,2024]

माना [t], t से कम या बराबर एक महत्तम पूर्णांक फलन है। माना A संख्या 2310 के सभी अभाज्य गुणनखण्डों को समुच्चय है और $f: A \to Z$ एक फलन $f(x)=\left[\log_2\left(x^2+\left[\frac{x^3}{5}\right]\right)\right]$ तो समुच्चय A से फलन f के परिसर में एकैकी फलनों की संख्या होगी :

[JEE(Main)April,2024]

(A) 25 (B) 20 (C) 120 (D) 24

Ans. C

Sol. $N = 2310 = 231 \times 10$

$= 3 \times 11 \times 7 \times 2 \times 5$

$A = \{2,3,5,7,11\}$

$$f(x)=\left[\log_2\left(x^2+\left[\frac{x^3}{5}\right]\right)\right]$$

$f(2) = [\log_2(5)] = 2$
$f(3) = [\log_2(25 + 25)] = 5$
$f(7) = [\log_2(117)] = 6$
$f(11) = [\log_2 387] = 8$
Range of f: $B = \{2,3,5,6,8\}$

No. of one-one functions = 5! = 120

41. Let $f(x)=\begin{cases} -a & \text{if } -a \le x \le 0 \\ x+a & \text{if } 0 < x \le a \end{cases}$ where $a > 0$ and $g(x)=\frac{\left(f(|x|)-|f(x)|\right)}{2}$. Then the function $g : [-a, a] \to [-a, a]$ is :

[JEE(Main)April,2024]

(A) One-one (B) Neither one-one nor onto

(C) Onto (D) Both one-one and onto

माना $f(x)=\begin{cases} -a & \text{if } -a \le x \le 0 \\ x+a & \text{if } 0 < x \le a \end{cases}$ है, जहाँ $a > 0$ तथा $g(x)=\frac{\left(f(|x|)-|f(x)|\right)}{2}$ है। तो फलन $g : [-a, a] \to [-a, a]$ होगा :

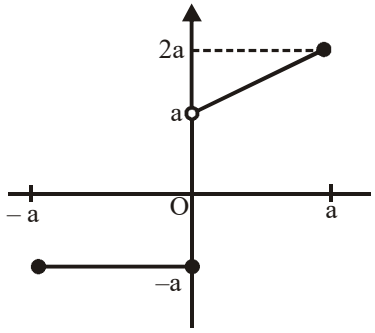
[JEE(Main)April,2024]

(A) एकैकी है (B) न तो एकैकी है न ही आच्छादक है

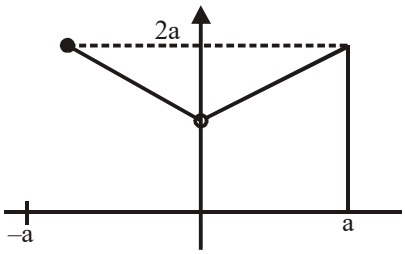
(C) आच्छादक है (D) ऐकैकी तथा आच्छादक दोनों है

Ans. B

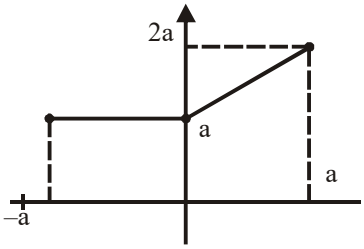
Sol. $y = f(x)$



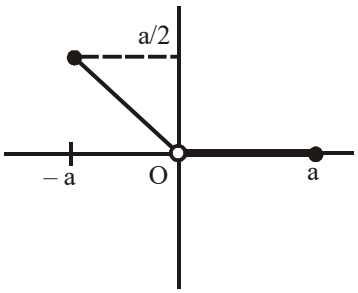
$y = f|x|$



$y = |f(x)|$



$$g(x) = \frac{f(|x|) - |f(x)|}{2}$$



42. Let $f(x) = x^2 + 9$, $g(x) = \frac{x}{x-9}$ and $a = fog(10)$, $b = gof(3)$. If e and l denote the eccentricity and the length of the latus rectum of the ellipse $\frac{x^2}{a} + \frac{y^2}{b} = 1$, then $8e^2 + l^2$ is equal to : **[JEE(Main)April,2024]**

माना $f(x)=x^2+9,\ g(x)=\frac{x}{x-9}$ है और $a=fog(10), b=gof(3)$ है। यदि दीर्घवृत्त $\frac{x^2}{a}+\frac{y^2}{b}=1$ की उत्केन्द्रता तथा नाभिलंब जीवा की लंबाई क्रमशः e तथा l है। तो $8e^2+l^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 6 (B) 12 (C) 16 (D) 8

Ans. D

Sol. $f(x)=x^2+9$

$$g(x)=\frac{x}{x-9}$$

$f(g(10))=f(10)=109=a$

$g(f(3))=2=b$

$$\frac{x^2}{109}+\frac{y^2}{2}=1$$

$2=109\,(1-e^2)$

$$e^2=\frac{107}{109}$$

$$\ell=\frac{2\times2}{\sqrt{109}}\Rightarrow\ell^2=\frac{16}{109}$$

$$8e^2+\ell^2=\frac{856}{109}+\frac{16}{109}=8$$

43. Let the range of the function $f(x)=\frac{1}{2+\sin 3x+\cos 3x}$, $x\in R$ be [a, b]. If α and β are respectively the A.M. and the G.M. of a and b, then $\frac{\alpha}{\beta}$ is equal to : **[JEE(Main)April,2024]**

माना फलन $f(x)=\frac{1}{2+\sin 3x+\cos 3x}$, $x\in R$ का परिसर [a, b] है। यदि a तथा b के A.M. तथा G.M. क्रमशः α तथा β हैं, तो $\frac{\alpha}{\beta}$ बराबर है : **[JEE(Main)April,2024]**

(A) $\sqrt{\pi}$ (B) π (C) 2 (D) $\sqrt{2}$

Ans. D

Sol. $$f(x)=\frac{1}{2+\sin 3x+\cos 3x}$$

we know that

$$-\sqrt{2}\le\sin 3x+\cos 3x\le\sqrt{2}$$

$$2-\sqrt{2} \le 2+\sin 3x+\cos 3x \le 2+\sqrt{2}$$

$$\frac{1}{2+\sqrt{2}} \le \frac{1}{2+\sin 3x+\cos 3x} \le \frac{1}{2-\sqrt{2}}$$

$$\therefore a = \frac{1}{2+\sqrt{2}} \text{ or } \frac{2-\sqrt{2}}{2}$$

$$b = \frac{1}{2-\sqrt{2}} \text{ or } \frac{2+\sqrt{2}}{2}$$

$$\alpha = \text{A. M of a and b} = \frac{\frac{2-\sqrt{2}}{2}+\frac{2+\sqrt{2}}{2}}{2} = 1$$

$$\beta = \text{G. M of a and b} = \sqrt{\frac{2-\sqrt{2}}{2}\times\frac{2+\sqrt{2}}{2}} = \frac{1}{\sqrt{2}}$$

$$\therefore \frac{\alpha}{\beta} = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2}$$

44. Consider the function f : R $\to$ R defined by $f(x)=\frac{2x}{\sqrt{1+9x^2}}$. If the composition of $f,\underbrace{(fofofo.....f)}_{10 \text{ times}}(x)=\frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$, then the vlaue of $\sqrt{3\alpha+1}$ is equal to _____.

[JEE(Main)April,2024]

विचार कीजिए फलन $f: R \to R$, $f(x)=\frac{2x}{\sqrt{1+9x^2}}$ द्वारा परिभाषित है। यदि f का संयोजन $f,\underbrace{(fofofo.....f)}_{10 \text{ times}}(x)=\frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$ है, तो $\sqrt{3\alpha+1}$ का मान बराबर है _____

[JEE(Main)April,2024]

Ans. 1024

Sol. $f(x)=\frac{2x}{\sqrt{1+9x^2}}$

$$(fof)(x)=\frac{2f(x)}{\sqrt{1+9(f(x))^2}}=\frac{\frac{4x}{\sqrt{1+9x^2}}}{\sqrt{1+9\times\frac{4x^2}{1+9x^2}}}=\frac{4x}{\sqrt{1+45x^2}}$$

$$(\text{fofof})(x)=\frac{4\times\frac{2x}{\sqrt{1+9x^2}}}{\sqrt{1+45\times\frac{4x^2}{1+9x^2}}}=\frac{8x}{\sqrt{1+21\times9x^2}}$$

$$(\text{fofofof})(x)=\frac{16x}{\sqrt{1+85\times9x^2}}$$

$\Rightarrow \alpha$ is 10th term of 1, 5, 21, 85, ...

α is 10th term of $\frac{\left(2^1\right)^2-1}{3},\frac{\left(2^2\right)^2-1}{3},\frac{\left(2^3\right)^2-1}{3},\frac{\left(2^4\right)^2-1}{3},\ldots$

$$\Rightarrow \alpha=\frac{\left(2^{10}\right)^2-1}{3}$$

$$\Rightarrow \sqrt{3\alpha+1}=2^{10}=1024$$

45. If $S=\left\{a\in R:|2a-1|=3[a]+2\{a\}\right\}$, where [t] deonts the greatest integer less than or equal to t and {t} represents the fractional part t, then $72\sum_{a\in S}$ a is equal to _____. **[JEE(Main)April,2024]**

यदि $S=\left\{a\in R:|2a-1|=3[a]+2\{a\}\right\}$ है, जहाँ [t] महत्तम पूर्णांक फलन है तथा {t}, t के भिन्नात्मक भाग को दर्शाता है, तो $72\sum_{a\in S}$ a बराबर है _____ **[JEE(Main)April,2024]**

Ans. 18

Sol. $|2a-1|=3[a]+2\{a\}$

Case-I : $a\geq\frac{1}{2}$

$2a-1=[a]+2a \qquad (\because\ a=[a]+\{a\})$

$[a]=-1$

$a\in[-1,0)$ but $a\geq\frac{1}{2}$, $\Rightarrow$ No value of a

Case-II: $a<\frac{1}{2}$

$1-2a=[a]+2a$

$1=[a]+4a$

$1=5[a]+4\{a\}$

$4\{a\}=1-5[a]$

$0\leq 4\{a\}<4$

$0 \le 1 - 5[a] < 4$

$\frac{1}{3} \ge [a] \ge -\frac{3}{5} \quad \Rightarrow \quad [a] = 0 \quad \Rightarrow \quad \{a\} = \frac{1}{4}$

$\Rightarrow \quad a = \frac{1}{4}$

Now, $72\sum a = 72 \times \frac{1}{4} = 18$

46. If a function f satisfies $f(m+n) = f(m) + f(n)$ for all $m, n \in N$ and $f(1) = 1$, then the largest natural number λ such that $\sum_{k=1}^{2022} f(\lambda + k) \le (2022)^2$ is equal to _____. **[JEE(Main)April,2024]**

यदि सभी $m, n \in N$ के लिए, $f(m + n) = f(m) + f(n)$ है तथा $f(1) = 1$ है, के अधिकतम धन पूर्णांक λ जिसके लिए $\sum_{k=1}^{2022} f(\lambda + k) \le (2022)^2$ है, है _____ **[JEE(Main)April,2024]**

Ans. 1010

Sol. $f(m+n) = f(m) + f(n)$

$\Rightarrow f(x) = kx$

$\Rightarrow f(1) = 1$

$\Rightarrow k = 1$

$f(x) = x$

Now

$\sum_{k=1}^{2022} f(\lambda + k) \le (2022)^2$

$\Rightarrow \sum_{k=1}^{2022} (\lambda + k) \le (2022)^2$

$\Rightarrow 2022\lambda + \frac{2022 \times 2023}{2} \le (2022)^2$

$\Rightarrow \lambda \le 2022 - \frac{2023}{2}$

$\Rightarrow \lambda \le 1010.5$

47. Let $A = \{(x, y) : 2x + 3y = 23, x, y \in N\}$ and $B = \{x : (x, y) \in A\}$. Then the number of one-one functions from A to B is equal to _____. **[JEE(Main)April,2024]**

माना $A = \{(x, y) : 2x + 3y = 23, x, y \in N\}$ तथा $B = \{x : (x, y) \in A\}$ हैं। तो A से B के एकैकी फलनों की संख्या है

_____.

Ans. 24

Sol. $2x + 3y = 23$

$x = 1 \quad y = 7$

$x = 4 \quad y = 5$

$x = 7 \quad y = 3$

$x = 10 \quad y = 1$

A	B
(1, 7)	1
(4, 5)	4
(7, 3)	7
(10, 1)	10

The number of one-one functions from A to B is equal to 4!